

Structural Constraints on Observable Algebras from Finite-Rank Non-Invertible Projections

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Abstract

Let $\Pi : \mathcal{H}_{\text{bulk}} \rightarrow \mathcal{H}_{\text{bdy}}$ be a bounded linear projection with non-trivial kernel and finite rank. Four structural consequences follow without assuming any specific dynamical law or field content.

The kernel $\ker \Pi$ is infinite-dimensional: for any system whose admissibility conditions suppress normal transport at the boundary, distinct bulk configurations generate identical boundary observables exactly. The mutual information between admissible bulk states and their Π -images is bounded above by a finite observability capacity $\mathcal{C}(\Gamma) < \infty$ depending only on the interface conditioning parameter Γ , independent of measurement strategy or inversion algorithm, and structurally analogous to Shannon's channel capacity. The absolute dimensional scale of the bulk belongs to $\ker(\Pi^*)$: no functional on the boundary algebra can recover it from the adimensional subalgebra alone, and a single external metrological anchor is necessary and sufficient to complete the dimensional structure. When no coherent mode survives the projection, the residual boundary correlations are scale-free, with power spectral density $\propto \omega^{-1}$; this $1/f$ floor is the irreducible observable signature of the kernel.

These four results are proved under four minimal hypotheses (H1)–(H4). They are derived first for wave systems at strongly asymmetric interfaces, where the kernel is explicitly constructed, and then extracted in model-independent form. Applications include finite-rank quantum systems, holographic models, and any theory whose observable sector arises from a finite-rank non-isometric projection of a bulk.

1 Introduction

When a physical system has a bulk description on a domain Ω and a boundary description on $\Sigma = \partial\Omega$, the passage from bulk to boundary is governed by a restriction map $P : \mathcal{H}_\Omega \rightarrow \mathcal{H}_\Sigma$. Whenever this map is non-invertible — which it must be when $\dim \mathcal{H}_\Sigma < \dim \mathcal{H}_\Omega$ — the observable algebra on the boundary is strictly contained in the full algebra of the bulk. The question is what follows structurally from this containment, independently of any specific dynamical law or field content.

At a strongly asymmetric interface, where the bulk supports modes that the boundary layer cannot sustain in propagating form, suppression of normal transport forces a non-invertible projection with an infinite-dimensional kernel. This is not a modelling assumption: it is a consequence of the admissibility conditions imposed by the interface. The

structural features that emerge in this setting — infinite-dimensional kernel, finite observability capacity, adimensional boundary algebra, $1/f$ noise residue — hold for any system satisfying four minimal hypotheses (H1)–(H4), stated in Section 6. The wave-physics case provides the technical core; Section 6 extracts the general form.

2 The kernel of the bulk-to-boundary map

2.1 Setting

Let $\Omega \subset \mathbb{R}^n$ be an open half-space with boundary $\Sigma = \{x : x_\xi = 0\}$, where ξ denotes the normal coordinate. We consider bulk fields $X : \Omega \rightarrow \mathbb{C}$ governed by a linear wave equation with constant coefficients:

$$\mathcal{L}X = 0 \quad \text{in } \Omega. \quad (1)$$

The bulk-to-boundary restriction map is:

$$P : X \mapsto X|_\Sigma. \quad (2)$$

A phase functional F extracts a scalar observable from the boundary trace: $M[X] = (F \circ P)[X]$.

Definition 2.1 (Admissible bulk field). A field X solving (1) is *admissible* if it satisfies the interface conditions on Σ : continuity of tangential field components and (where relevant) traction-free or impedance-matching conditions. The space of admissible fields is $\mathcal{A} \subset L^2(\Omega)$.

The key structural condition is the suppression of normal transport.

Definition 2.2 (Normal transport suppression). An interface Σ exhibits *normal transport suppression* (NTS) if the admissibility conditions imply that the normal energy flux $\langle J_\xi \rangle_\Sigma = 0$ for all fields in $\mathcal{A}$.

NTS holds at any interface where the two media support fundamentally different admissible wave dynamics: electromagnetic waves in air versus elastic waves in a solid, near-field evanescent modes versus propagating modes in the far field. Under NTS, propagating normal transport is excluded for the boundary-layer contribution.

2.2 Non-invertibility theorem

Theorem 2.3 (Structural non-invertibility). *Let Σ satisfy NTS (Definition 2.2). Then the restriction map $P : \mathcal{A} \rightarrow L^2(\Sigma)$ is not injective. Moreover, $\ker P$ is infinite-dimensional: it contains a distinct non-trivial element for each pair $(\mathbf{k}_\parallel, \omega)$ with $\mathbf{k}_\parallel \neq \mathbf{0}$ and $\omega > 0$.*

Proof. Fix $\mathbf{k}_\parallel \neq \mathbf{0}$ and $\omega > 0$. Let $k_\xi^{(1)} > 0$ and $k_\xi^{(2)} = -k_\xi^{(1)}$ be two real normal wave numbers satisfying the dispersion relation $\omega^2/c^2 = k_\xi^2 + |\mathbf{k}_\parallel|^2$. Define the standing-wave combination:

$$Z(\mathbf{x}, t) = (e^{ik_\xi^{(1)}x_\xi} - e^{-ik_\xi^{(1)}x_\xi}) e^{i(\mathbf{k}_\parallel \cdot \mathbf{x}_\parallel - \omega t)}. \quad (3)$$

By construction:

1. $\mathcal{L}Z = 0$: Z satisfies the wave equation.
2. $Z|_{x_\xi=0} = 0$: the two components cancel at the boundary, so $PZ = 0$.
3. $\|Z\|_{L^2(\Omega)} \neq 0$: Z is non-trivial in the bulk.
4. $Z \in \mathcal{A}$: the normal flux of Z at Σ vanishes by symmetry of the two components, so the NTS condition is satisfied.

Since the construction is valid independently for each $(\mathbf{k}_\parallel, \omega)$, the elements $\{Z_{\mathbf{k}_\parallel, \omega}\}$ are linearly independent and all lie in $\ker P$, making $\ker P$ infinite-dimensional. $\square$ $\square$

Remark 2.4 (Physical interpretation). The kernel element Z is a standing wave in the normal direction: the superposition of two counter-propagating modes that cancel exactly at Σ , while sustaining $O(1)$ energy in the bulk. Bulk energy can be stored in this configuration without any trace at the boundary. This is not a pathological special case: it is a generic feature of any interface where the dispersion relation admits two real normal wave numbers.

Corollary 2.5 (Hard ceiling on inversion). *For any measurement functional $M = F \circ P$ and any two admissible bulk configurations X_1, X_2 with $X_2 - X_1 \in \ker P$:*

$$M[X_1] = M[X_2] \quad (4)$$

exactly, in the noiseless limit, for any choice of F . No inversion algorithm can distinguish X_1 from X_2 using boundary data alone.

Proof. $M[X_2] = F(PX_2) = F(PX_1 + P(X_2 - X_1)) = F(PX_1) = M[X_1]$, since $P(X_2 - X_1) = 0$ by assumption. $\square$ $\square$

Numerical verification. With parameters $c = 1$, $\omega = 2\pi$, $k_\parallel = 0.6\omega/c$, the bulk difference $\|X_2 - X_1\|_{\text{bulk}}$ is of order unity ($\approx 0.57\|X_1\|$), while the maximum boundary discrepancy $\max_{\mathbf{x}_\parallel, t} |PX_2 - PX_1|$ and the phase discrepancy $\max_{\mathbf{x}_\parallel, t} |M[X_2] - M[X_1]|$ are both zero to floating-point precision.

The bulk decomposes orthogonally as:

$$\mathcal{A} = \ker P \oplus (\ker P)^\perp, \quad (5)$$

and the boundary observes only the $(\ker P)^\perp$ component.

3 The boundary observability capacity

Theorem 2.3 characterises what is lost. The complementary question is how much information about the bulk can any boundary observer recover.

3.1 Definition and structure

Definition 3.1 (Interface conditioning parameter). The *conditioning parameter* $\Gamma(\omega)$ is the ratio of the characteristic impedances of the two media at frequency ω : $\Gamma(\omega) = Z_1(\omega)/Z_2(\omega)$. It measures the degree of interface asymmetry. The two degenerate limits are: $\Gamma \rightarrow 0$ (one medium acoustically rigid), $\Gamma \rightarrow \infty$ (one medium electromagnetically opaque).

Definition 3.2 (Boundary observability capacity). The *boundary observability capacity* $\mathcal{C}(\Gamma)$ is:

$$\mathcal{C}(\Gamma) = \sup_{p(X) \in \mathcal{P}(\mathcal{A})} I(X; PX), \quad (6)$$

where $\mathcal{P}(\mathcal{A})$ is the set of probability distributions over admissible bulk configurations, and $I(X; PX)$ is the Shannon mutual information between the bulk configuration X and its boundary projection PX .

Theorem 3.3 (Boundary observability capacity). *Under NTS (Definition 2.2), the capacity $\mathcal{C}(\Gamma)$ satisfies:*

- (i) $\mathcal{C}(\Gamma) \rightarrow 0$ as $\Gamma \rightarrow 0$ (*elastic-dominated: no phase information crosses the boundary*);
- (ii) $\mathcal{C}(\Gamma) \rightarrow 0$ as $\Gamma \rightarrow \infty$ (*EM-dominated: the boundary trace is phase-insensitive*);
- (iii) $\mathcal{C}(\Gamma)$ attains a strictly positive maximum at some $\Gamma_k \in (0, \infty)$, the optimal conditioning.

The bound $\mathcal{C}(\Gamma)$ is independent of measurement strategy, sensor count, and inversion algorithm.

Proof sketch. The mutual information $I(X; PX)$ measures the statistical dependence between X and its projection PX . Under NTS, P suppresses all modes with non-zero normal flux; the image of P therefore has finite effective rank. In the limit $\Gamma \rightarrow 0$, the interface becomes acoustically rigid: boundary conditions force $PX = 0$ for all $X \in \mathcal{A}$, so $I(X; PX) = 0$. In the limit $\Gamma \rightarrow \infty$, the interface becomes electromagnetically opaque: all phase information is confined to the bulk, and $I(X; PX) = 0$. At intermediate Γ , a finite fraction of the $(\ker P)^\perp$ sector is accessible; the maximum over $p(X)$ is attained by the uniform distribution over this sector (maximum entropy input). The optimal Γ_k minimises the projection loss; it depends only on the impedance ratio and is independent of the input distribution. $\square$ $\square$

3.2 Structural analogy with Shannon capacity

The structure of $\mathcal{C}(\Gamma)$ mirrors Shannon's channel capacity $C(\text{SNR}) = \log_2(1 + \text{SNR})$: In Shannon's theorem, channel capacity depends on the channel, not on the message. Here, $\mathcal{C}(\Gamma)$ depends on the interface geometry, not on the measurement strategy. Both are hard upper bounds: no technique can exceed them. The optimal conditioning Γ_k is the interface analogue of the optimal operating SNR: it minimises projection loss.

4 The adjoint kernel and the metrological anchor

A third question differs in character: is there a feature of the bulk that is inaccessible to the boundary algebra regardless of its richness? The answer is yes. The absolute dimensional scale belongs to $\ker(\Pi^*)$.

Definition 4.1 (Adimensional subalgebra). The *adimensional subalgebra* $\mathcal{O}_{\text{bdy}}^{(0)} \subset \mathcal{O}_{\text{bdy}}$ is the subalgebra generated by all dimensionless combinations of projected bulk quantities: ratios, products, and scalar functions thereof.

Under finite-rank projection, $\mathcal{O}_{\text{bdy}}$ is generated by a finite set of modes. Their ratios and products are dimensionless. Therefore $\mathcal{O}_{\text{bdy}}^{(0)}$ is the natural and complete object of the boundary theory: it encodes everything the boundary can say about the shape of the bulk, independently of any choice of units.

Theorem 4.2 (Absolute scale in $\ker(\Pi^*)$). Let $\Pi : \mathcal{H}_{\text{bulk}} \rightarrow \mathcal{H}_{\text{bdy}}$ be a bounded linear projection with $\ker \Pi \neq \{0\}$. Let $\mathcal{O}_{\text{bdy}}$ be the algebra generated by $\text{Im}(\Pi)$, and let ℓ_{abs} be any quantity that sets the dimensional unit of $\mathcal{H}_{\text{bulk}}$ (a length, a frequency, or a mass).

Then no bounded linear functional $\phi : \mathcal{O}_{\text{bdy}} \rightarrow \mathbb{R}$ can recover ℓ_{abs} from $\mathcal{O}_{\text{bdy}}^{(0)}$ alone. Equivalently, ℓ_{abs} belongs to $\ker(\Pi^*)$, the kernel of the adjoint projection.

Proof. Let $\lambda > 0$ and consider the rescaling $\ell_{\text{abs}} \rightarrow \lambda \ell_{\text{abs}}$ of the bulk dimensional unit. Under this rescaling, all quantities in $\mathcal{H}_{\text{bulk}}$ scale by appropriate powers of λ , but all dimensionless ratios in $\mathcal{O}_{\text{bdy}}^{(0)}$ are invariant.

Suppose for contradiction that there exists a bounded functional $\phi : \mathcal{O}_{\text{bdy}} \rightarrow \mathbb{R}$ with $\phi(A) = \ell_{\text{abs}}$ for some $A \in \mathcal{O}_{\text{bdy}}^{(0)}$. Since A is dimensionless, its value is unchanged by the rescaling: $A \rightarrow A$ as $\ell_{\text{abs}} \rightarrow \lambda \ell_{\text{abs}}$. But then $\phi(A) = \ell_{\text{abs}}$ and $\phi(A) = \lambda \ell_{\text{abs}}$ simultaneously, which is impossible for $\lambda \neq 1$. Contradiction.

In the language of the adjoint: the inner product $\langle \Pi^* f, \ell_{\text{abs}} \rangle_{\mathcal{H}_{\text{bulk}}} = 0$ for all $f \in \mathcal{H}_{\text{bdy}}^*$, since ℓ_{abs} is in the scaling direction that Π annihilates in the adimensional sense. Therefore $\ell_{\text{abs}} \in \ker(\Pi^*)$. $\square$ $\square$

Remark 4.3 (Interpretation). Theorem 4.2 should not be read as a failure of the boundary description. It is a precise statement about the intrinsic structure of $\mathcal{O}_{\text{bdy}}$: the boundary algebra is *adimensional by necessity*, not by choice. It determines the shape of the world (all dimensionless relations) but not its absolute scale (the unit of measurement). That scale requires exactly one external input.

Definition 4.4 (Metrological anchor). A *metrological anchor* is a dimensional quantity r_{ext} that is:

- (a) *accessible*: r_{ext} is an observable quantity in the boundary sector (it exists inside the projected world);
- (b) *external to $\mathcal{O}_{\text{bdy}}^{(0)}$* : r_{ext} carries physical units that cannot be expressed as a ratio or product of elements of the adimensional subalgebra.

The anchor is a single measured quantity, internal to the boundary but external to the dimensional closure of $\mathcal{O}_{\text{bdy}}^{(0)}$.

Proposition 4.5 (Necessity and sufficiency of a single anchor). *Under the hypotheses of Theorem 4.2: At least one anchor is required (necessity: Theorem 4.2). A single anchor r_{ext} suffices (sufficiency): every dimensional observable $\mathcal{Q}$ writes as $r_{\text{ext}}^{d_{\mathcal{Q}}} \cdot q$ with $q \in \mathcal{O}_{\text{bdy}}^{(0)}$, so*

$$\mathcal{O}_{\text{bdy}}^{(0)} \cup \{r_{\text{ext}}\} \vdash \text{all dimensional observables.} \quad (7)$$

Proof. Necessity follows from Theorem 4.2. Sufficiency follows because any $\mathcal{Q}$ with $d_{\mathcal{Q}} \neq 0$ writes as $r_{\text{ext}}^{d_{\mathcal{Q}}} \cdot q$ with $q \in \mathcal{O}_{\text{bdy}}^{(0)}$ determined by the adimensional algebra; no second anchor is needed. $\square$ $\square$

Corollary 4.6 (Uniqueness of the anchor structure). *Any two anchors r_{ext} and r'_{ext} that satisfy Definition 4.4 are related by a dimensionless factor $\lambda \in \mathcal{O}_{\text{bdy}}^{(0)}$: $r'_{\text{ext}} = \lambda \cdot r_{\text{ext}}$. The anchor is unique up to the freedom in $\mathcal{O}_{\text{bdy}}^{(0)}$.*

5 The scale-free residue of the kernel

The kernel $\ker P$ is not merely missing information. Its presence generates a measurable signature at the boundary: a scale-free correlation structure.

Proposition 5.1 (Two limits of the projection). *Let Π be a finite-rank non-invertible projection.*

- (i) *When exactly one mode from $(\ker P)^\perp$ survives, the boundary observable is a structural invariant: a dimensionless constant determined solely by the projection geometry.*
- (ii) *When all modes in $(\ker P)^\perp$ are eliminated prior to observation, the residual boundary correlations are scale-free: $S(\omega) \propto \omega^{-1}$.*

Proof. (i) If a single mode $\psi_0 \in (\ker P)^\perp$ survives, the boundary observable is $M = F(P\psi_0)$. Since ψ_0 is the dominant mode of Π in $(\ker P)^\perp$ and F is fixed by the measurement, M depends only on the operator and not on any free parameter. It is a structural invariant.

(ii) When all coherent degrees of freedom are eliminated, no privileged frequency scale is available and the correlations are constrained to be scale-invariant by the symmetry of the incoherent limit. Scale invariance of $C(t) = C(\lambda t)$ for all $\lambda > 0$ implies $C(t) \propto t^{-1}$, and by Wiener-Khinchin, $S(\omega) \propto \omega^{-1}$. $\square$ $\square$

Remark 5.2 ($1/f$ as kernel signature). The $1/f$ floor is the irreducible observable signature of the kernel. It is not engineering noise: it is structural. It encodes information about what did not pass — the kernel — in the statistics of what did. The presence of $1/f$ noise at the boundary is diagnostic of an underlying non-invertible projection; its absence implies that at least one coherent mode has survived.

The two limits of Proposition 5.1 are connected by a continuous decoherence parameter $\sigma \in [0, 1]$, where $\sigma = 0$ is the fully coherent limit and $\sigma = 1$ is the fully incoherent limit. The observable capacity interpolates:

$$\mathcal{C}_\sigma(\Gamma) = (1 - \sigma)\mathcal{C}(\Gamma) + \sigma \cdot 0 = (1 - \sigma)\mathcal{C}(\Gamma), \quad (8)$$

and the power spectral density interpolates between the coherent (discrete spectrum) and incoherent ($1/f$) limits.

The two limits are therefore not different phenomena. They are the two faces of the same projection, seen from opposite ends of the decoherence spectrum.

6 The model-independent framework

(H1) There exists a separable Hilbert space $\mathcal{H}_{\text{bulk}}$ of bulk configurations.

(H2) There exists a bounded linear map $\Pi : \mathcal{H}_{\text{bulk}} \rightarrow \mathcal{H}_{\text{bdy}}$ with $\|\Pi\| \leq 1$ and $\ker \Pi \neq \{0\}$.

(H3) The observable algebra $\mathcal{O}_{\text{bdy}}$ is the C^* -algebra generated by $\{\Pi X \cdot \Pi Y^* : X, Y \in \mathcal{H}_{\text{bulk}}\}$.

(H4) $\text{rank}(\Pi) < \infty$.

These hypotheses are satisfied by: wave fields at asymmetric interfaces (Sections 2–5); finite-rank quantum channels; holographic dualities where the boundary algebra has finite central charge; and physical theories whose observable sector arises from integration over compact internal degrees of freedom.

Remark 6.1. Hypothesis (H4) concerns the image of Π , not its kernel. A finite-rank projection $\Pi : \mathcal{H}_{\text{bulk}} \rightarrow \mathcal{H}_{\text{bdy}}$ with $\dim \mathcal{H}_{\text{bulk}} = \infty$ necessarily has $\dim \ker \Pi = \infty$ by the rank-nullity theorem. The finite-dimensional image and the infinite-dimensional kernel are thus consistent and both required for Theorem 6.2.

Theorem 6.2 (Structural constraints). *Under (H1)–(H4), the following hold simultaneously:*

(i) *$\ker \Pi$ is non-trivial. Distinct bulk configurations generate identical observables in $\mathcal{O}_{\text{bdy}}$.*

(ii) *The mutual information between bulk states and their Π -images is bounded: $\sup_{p(X)} I(X; \Pi X) < \infty$.*

(iii) The adimensional subalgebra $\mathcal{O}_{\text{bdy}}^{(0)} \subsetneq \mathcal{O}_{\text{bdy}}$ does not determine the absolute dimensional scale of $\mathcal{H}_{\text{bulk}}$. A single metrological anchor is necessary and sufficient to complete the dimensional structure.

(iv) In the fully incoherent limit (all modes in $(\ker \Pi)^\perp$ eliminated), the residual correlations in $\mathcal{O}_{\text{bdy}}$ are scale-free.

Proof. (i) Hypothesis (H2) gives $\ker \Pi \neq \{0\}$ directly. (ii) Finite rank (H4) implies that the image of Π lies in a finite-dimensional subspace, bounding the information capacity. (iii) Theorem 4.2 and Proposition 4.5 hold under (H1)–(H3); they do not require (H4). (iv) Proposition 5.1(ii) holds for any non-invertible Π under the incoherent limit. $\square$ $\square$

Hypothesis (H4) implies that $\mathcal{O}_{\text{bdy}}^{(0)}$ is complete in the following sense.

Corollary 6.3 (Completeness of the adimensional algebra). *Under (H1)–(H4), the adimensional subalgebra $\mathcal{O}_{\text{bdy}}^{(0)}$, together with a single metrological anchor r_{ext} , determines all physical observables of the system:*

$$\mathcal{O}_{\text{bdy}}^{(0)} \cup \{r_{\text{ext}}\} \vdash \text{all observables.} \quad (9)$$

No additional input is required or consistent with the structure of Π .

The boundary algebra determines everything about the bulk that is dimensionless. One anchor provides the rest.

The four results fit into a precise hierarchy:

$$\mathcal{H}_{\text{bulk}} \xrightarrow{\Pi} \mathcal{O}_{\text{bdy}} \supset \mathcal{O}_{\text{bdy}}^{(0)} \xleftarrow{\cup} \{r_{\text{ext}}\} \vdash \text{all observables.} \quad (10)$$

The bulk maps to $\mathcal{O}_{\text{bdy}}$ under Π ; the adimensional subalgebra encodes shape without scale; one anchor completes the dimensional structure. The kernel $\ker \Pi$ is the complement: bulk content that does not appear in $\mathcal{O}_{\text{bdy}}$ under any strategy, with the $1/f$ residue as its observable signature.

7 Conclusions

Under hypotheses (H1)–(H4), a finite-rank non-invertible projection $\Pi : \mathcal{H}_{\text{bulk}} \rightarrow \mathcal{H}_{\text{bdy}}$ forces four structural consequences on the observable algebra, independent of the specific dynamics or field content. The kernel $\ker \Pi$ is infinite-dimensional: distinct bulk configurations are observationally identical at the boundary, and no inversion strategy removes this degeneracy. The recoverable bulk information is bounded above by a finite observability capacity $\mathcal{C}(\Gamma)$ depending only on the interface geometry, structurally analogous to Shannon’s channel capacity. The absolute dimensional scale of the bulk belongs to $\ker(\Pi^*)$: the boundary algebra is adimensional by necessity, and a single metrological anchor is necessary and sufficient to determine all dimensional observables. In the fully incoherent limit the residual boundary correlations are scale-free, $S(\omega) \propto \omega^{-1}$; this $1/f$ floor is the observable signature of the kernel.

The hierarchy (10) summarises the structure: the bulk maps to the boundary through irreversible loss; the adimensional algebra encodes the shape; one anchor fixes the scale. What does not pass is not absent — it is encoded in the $1/f$ residue.

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